Logging and tracing for Streamlit in Snowflake

Streamlit in Snowflake supports logging for both warehouse and container runtimes. Warehouse runtimes use the Snowflake telemetry framework to capture log messages and trace events into an event table. Container runtimes capture logs that your app emits to standard output and standard error, store them in the account's event table, and provide both live console logs and historical log views in Snowsight.

Both runtimes store logs in the account-level event table. An account administrator must set up and configure this event table before logs can be captured. For instructions, see Event table overview.

- To find the event table configured for your account, run:

```
SHOW PARAMETERS LIKE 'event_table' IN ACCOUNT;
```

The following table compares logging and tracing support by runtime:

Feature	Warehouse runtime	Container runtime
Event table logging	Supported	Supported
Tracing	Supported	Not supported
Live console logs in Snowsight	Not supported	Supported
Historical logs in Snowsight	Not s	

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Container runtime

FEATURE — GENERALLY AVAILABLE

Not supported in government regions.

Container-runtime Streamlit apps run inside a Snowpark Container Services container. Snowflake automatically captures anything your app emits to standard output and standard error and stores it in the account's event table. You can view these logs in Snowsight or query them with SQL.

Python's logging module

Use Python's built-in `logging` module to emit log messages from your app. The following example configures a logger that writes INFO-level and higher messages to standard output:

```
import logging
import sys

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s %(levelname)s %(name)s: %(message)s",
    stream=sys.stdout,
)

LOGGER = logging.getLogger("my_app")
```

In order from least to most severe, Python has the following logging levels:

- DEBUG
- INFO
- WARNING
- ERROR

Setting the level to INFO captures
messages.

Note

By default, Python's `logging` m
captures both standard output
the stream you use. Setting the
because standard error is conv

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After configuring the logger, you can use it to log messages throughout your app code. It is common to define a logger in a separate module and then import it into your app code:

```
source_directory/
├── my_logger.py
├── pyproject.toml
└── streamlit_app.py
```

```python
import streamlit as st
from my_logger import LOGGER

LOGGER.info("Home page loaded")
st.title("My App")

if st.button("Run analysis"):
    LOGGER.info("Analysis button clicked")
    try:
        result = run_analysis()
        LOGGER.info("Analysis completed successfully")
    except Exception as e:
        LOGGER.error("Analysis failed: %s", e)
        st.error("Analysis failed: %s", e)
```

Live logs in Snowsight

When you edit a container-runtime app in Snowsight, a logs pane appears below the editor. This pane streams log messages in real time as your app emits them. A short history of the most recent logs is displayed when you first connect.

Each log entry shows the following information:

Column	Description
`Source`	`APP` for logs from yo logs from the system
`Level`	The severity level of
`Message`	The log message co

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Available live-log actions

In the upper-right corner of the logs pane, you can search and filter the logs to help you find the information you need. This includes text search, filtering by source, and filtering by severity level. In the three-dot menu, you can download the current logs, navigate to the historical logs, or clear the live-logging pane. When you clear the pane, the current logs are deleted from your current view but not from the event table. Immediately reloading the page restores the most recent logs.

Understanding log sources

Logs from container-runtime apps have one of two sources:

- `MANAGER`: The system process inside the container that prepares and runs your app. Manager logs include messages about downloading your app files from the stage, installing Python dependencies, and starting the Streamlit server process. If you update your app's dependency files while the app is running, the manager process reinstalls dependencies and produces additional manager logs.
- `APP`: Logs from the running Streamlit server process. This includes messages from your user-configured Python loggers, Streamlit's built-in logger, and any other output your app writes to standard output or standard error.

The boundary between sources is the `streamlit run` command. Everything the container does before starting the Streamlit process produces `MANAGER` logs. After the Streamlit process starts, output from that process produces `APP` logs.

View historical logs in Snowsight

The following steps only apply to container-runtime apps. Warehouse runtime apps don't have a logs pane.

1. Sign in to Snowsight.
2. In the navigation menu, select
3. In the upper-right corner of th
4. In the upper-right corner of th **Historical logs**.

This opens the Snowpark Contain your app. The logs table shows th

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Column	Description
Timestamp	The timestamp of the log message.
Instance ID	The identifier for the container instance. This is always `0` for Streamlit apps.
Container	The identifier for the container instance.
Stream	Whether the log was emitted to standard output (`stdout`) or standard error (`stderr`).
Value	The JSON-formatted log message that includes `"level"`, `"message"`, `"source"`, and `"timestamp"` fields.

For more information about the monitoring page, see Snowpark Container Services: Monitoring Services.

Query logs with SQL

You can query the event table directly to analyze your container-runtime app's logs. The following query retrieves logs from a specific Streamlit app:

```
SELECT
    TIMESTAMP,
    RECORD['severity_text']::VARCHAR AS level,
    VALUE::VARCHAR AS message,
    RESOURCE_ATTRIBUTES['snow.database.name']::VARCHAR AS database_name,
    RESOURCE_ATTRIBUTES['snow.schema.name']::VARCHAR AS schema_name,
    RESOURCE_ATTRIBUTES['snow.executable.name']::VARCHAR AS app_name,
    RECORD_ATTRIBUTES['log.iostream']::VARCHAR AS stream
FROM <event_table>
WHERE RESOURCE_ATTRIBUTES['snow.database.name'] = '<database_name>'
  AND RESOURCE_ATTRIBUTES['snow.schema.name'] = '<schema_name>'
  AND RESOURCE_ATTRIBUTES['snow.executable.name'] = '<app_name>'
  AND RECORD_TYPE = 'LOG'
  AND TIMESTAMP > DATEADD
```

Replace `<event_table>` with the e
command, and replace `<database`
your Streamlit app.

Tip
Include a TIMESTAMP filter in y
tables can contain a large volu

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For more information about the event table columns, see Event table columns.

Warehouse runtime logging

For Streamlit apps using warehouse runtimes, you can capture log messages and trace events of your Streamlit app code as it runs and then analyze the results with SQL, for example, to analyze errors. For more information, see Logging, tracing, and metrics.

Warehouse runtimes require log and trace levels to be set on the database containing your app:

```
-- Set the log level for the database containing your app
ALTER DATABASE <database_name> SET LOG_LEVEL = INFO;

-- Set the trace level for the database containing your app
ALTER DATABASE <database_name> SET TRACE_LEVEL = ON_EVENT;
```

Example: Logging from a warehouse-runtime app

```
import logging
import streamlit as st

logger = logging.getLogger("simple_logger")

# Write directly to the app
st.title("Simple Logging Example")

# Get the current credentials
session = st.connection('snowflake').session()

def get_log_messages_query():
    return """
        SELECT
```


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Tracing (warehou

Tracing is supported for warehous
app and then query the event tabl

Note

The following example requires installing the `snowflake-telemetry-python` package. For more information, see Adding support for the telemetry package.

```
import streamlit as st
import time
import random
from snowflake import telemetry

def sleep_function() -> int:
    random_time = random.randint(1, 10)
    time.sleep(random_time)
    return random_time

def get_trace_messages_query() -> str:
    return """
            SELECT
                TIMESTAMP
```


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